Sharp Harmonic Cycle-Length Inequalities for Complete Multipartite Graphs

With bounded-cover and near-balanced extensions

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Abstract

For a finite simple graph G, let $\mathcal{C}(G)$ be the set of distinct lengths of its simple cycles and let $\mathsf{s}(G) = \sum_{\ell \in \mathcal{C}(G)} \ell^{-1}$. A conjecture of Erdős asserts that if $n \geq 2d$ and $e(G) \geq d(n-d)$, then $\mathsf{s}(G) \geq \frac{1}{2}(H_d - 1)$, with equality only for $K_{d,n-d}$. We prove a sharp version for every complete multipartite graph, for every integer $d \geq 2$, under the weaker hypothesis $e(G) > (d-1)(n-d+1)$. The proof is based on an exact classification of the cycle-length spectrum of complete multipartite graphs. We also establish the original inequality, with its equality characterization, for graphs having a vertex cover of size at most d under an explicit order hypothesis, and for arbitrary graphs in the near-balanced range $d \geq 4$, $2d \leq n \leq 2d+4$. The latter proof invokes classical theorems of Brandt, Erdős–Gallai, and Moon–Moser. These are restricted-family results, not a proof of the unrestricted conjecture.

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1 Introduction

We work throughout with finite, undirected, simple graphs. For a graph G of order n, define

$$\mathcal{C}(G) = \{\ell \in \{3, \dots, n\} : G \text{ contains a cycle of length } \ell\}, \qquad \mathsf{s}(G) = \sum_{\ell \in \mathcal{C}(G)} \frac{1}{\ell}.$$

Write $H_t = \sum_{j=1}^{t} j^{-1}$ for the tth harmonic number and

$$B_d = \sum_{j=2}^{d} \frac{1}{2j} = \frac{1}{2}(H_d - 1).$$

The complete bipartite graph $K_{d,n-d}$ has $\mathcal{C}(K_{d,n-d}) = \{4, 6, \dots, 2d\}$ whenever $n \geq 2d$; hence $\mathsf{s}(K_{d,n-d}) = B_d$.

A sharp form of the Erdős–Hajnal harmonic cycle-length problem asks whether

$$n \geq 2d, \quad e(G) \geq d(n-d) \qquad \Longrightarrow \qquad \mathsf{s}(G) \geq B_d, \tag{1}$$

with equality only when $G \cong K_{d,n-d}$; see [1, 2]. Montgomery, Milojević, Pokrovskiy, and Sudakov [1] recently established the stronger edge-threshold version for all sufficiently large d. It is important that this known theorem does not state a uniform result for every $d \geq 2$.

Our main result proves their sharp edge-threshold inequality for *all* $d \geq 2$ under the complete multipartite hypothesis. We give an exact cycle-spectrum formula rather than assuming a general pancyclicity theorem. Two additional results concern graphs whose structure is less rigid: one has an unbounded number of vertices but a small vertex cover; the other applies to arbitrary graphs in a short near-balanced interval of orders.

Theorem 1.1 (Sharp complete multipartite case)**.** *Let $d \geq 2$ and $n \geq 2d$ be integers. If G is a complete multipartite graph on n vertices and*

$$e(G) > (d-1)(n-d+1), \tag{2}$$

then

$$\mathsf{s}(G) \geq B_d.$$

Equality holds if and only if $G \cong K_{d,n-d}$. The strict threshold in (2) cannot be weakened to a non-strict one, even within complete multipartite graphs.

The two extensions are as follows.

Theorem 1.2 (Bounded vertex cover)**.** *Let $d \geq 2$ and $n \geq 2d + \binom{d}{2}$. If an n-vertex graph G satisfies $\tau(G) \leq d$ and $e(G) \geq d(n-d)$, then $\mathsf{s}(G) \geq B_d$, with equality if and only if $G \cong K_{d,n-d}$.*

Here $\tau(G)$ denotes the minimum size of a vertex cover.

Theorem 1.3 (Arbitrary graphs in a near-balanced range)**.** *If $d \geq 4$ and $2d \leq n \leq 2d+4$, then every n-vertex graph G with $e(G) \geq d(n-d)$ satisfies $\mathsf{s}(G) \geq B_d$, with equality if and only if $G \cong K_{d,n-d}$.*

2 The complete multipartite cycle spectrum

We start with a precise classification that supplies the principal combinatorial input. The graph G is *complete multipartite* if its vertex set is a partition into nonempty independent classes, and two vertices are adjacent precisely when they belong to different classes.

Lemma 2.1 (Capped selection)**.** *Let $a_1, \dots, a_r$ be positive integers, $r \geq 3$, with sum n, and put $b = \max_i a_i$. Suppose $b < n/2$. Then for every $3 \leq \ell \leq n$, with $q = \lfloor \ell/2 \rfloor$, one has*

$$\sum_{i=1}^{r} \min(a_i, q) \geq \ell. \tag{3}$$

Proof. If all $a_i \leq q$, the left side equals $n \geq \ell$. If exactly one class has size greater than q, that class contributes q. The other classes together have $n - b > b > q$ vertices, so they contribute at least $q+1$; here their sizes are individually at most q. Thus the sum is at least $2q+1 \geq \ell$. Finally, if two or more classes exceed q, their combined contribution is at least $2q$. This is enough when $\ell = 2q$. When $\ell = 2q+1$, there is at least one further nonempty class, unless three or more already exceed q, in either event contributing at least one extra unit. This proves (3). □

Proposition 2.2 (Exact cycle-length spectrum)**.** *Let G be complete multipartite, with $r \geq 2$ nonempty classes, n vertices, largest class size b, and $s = n - b$.*

(i) If $r = 2$, then $\mathcal{C}(G) = \{4, 6, \dots, 2s\}$.

(ii) If $r \geq 3$, then

$$\mathcal{C}(G) = \{3, 4, \dots, \min(n, 2s)\}. \tag{4}$$

Proof. For (i), $G = K_{b,s}$ with $b \geq s$, so the alternating cycles have exactly the lengths displayed, and a bipartite graph has no odd cycle.

For (ii), first suppose $b \geq s$. There are at least two classes outside the largest class, so $s \geq 2$. For $2 \leq q \leq s$ select q vertices in the largest class and q vertices outside it. Their complete bipartite cross-edges contain a $2q$-cycle. For $1 \leq q \leq s-1$, select q vertices in the largest class and $q+1$ outside vertices in such a way that no outside class contributes more than q vertices. This is possible because the outside classes have total size $s \geq q+1$ and at least two are nonempty: if one exceeds q, the others contribute at least one; otherwise all s vertices are eligible. The selected $(2q+1)$-vertex induced graph has every class of size at most q and hence minimum degree at least $(2q+1)-q = q+1$. Dirac's Hamiltonicity theorem [3] (also valid for order three) supplies a cycle of length $2q+1$. All lengths $3, 4, \ldots, 2s$ occur.

No cycle can use more than s vertices from the largest class: vertices of that class are independent, and successive such vertices around a cycle must be separated by distinct vertices outside the class. Therefore every cycle has length at most $2s$. As $b \geq s$, $\min(n, 2s) = 2s$, proving (4) in this case.

Now suppose $b < s$, or equivalently $b < n/2$. Fix $3 \leq \ell \leq n$ and let $q = \lfloor \ell/2 \rfloor$. Lemma 2.1 allows the selection of exactly ℓ vertices, with at most q from any class. The induced complete multipartite graph has minimum degree at least $\ell - q = \lceil \ell/2 \rceil$. Dirac's theorem therefore supplies an ℓ-cycle. Every length from 3 through n occurs, completing the proof. □

Corollary 2.3 (Exact harmonic sum)**.** *With the notation of Proposition 2.2,*

$$\mathsf{s}(G) = \begin{cases} \frac{1}{2}(H_s - 1), & r = 2, \\ H_{\min(n,2s)} - \frac{3}{2}, & r \geq 3. \end{cases}$$

Proof. Sum the reciprocals of the lengths listed in Proposition 2.2. □

3 Proof of the sharp multipartite theorem

Proof of Theorem 1.1. Let the class sizes of G be $a_1 \geq \cdots \geq a_r > 0$, where $r \geq 2$, $b = a_1$, and $s = n - b$. The case $r = 1$ is ruled out by (2).

Suppose first that $r = 2$. The graph is $K_{s,n-s}$, with $s \leq n/2$ and $e(G) = s(n-s)$. The function $x \mapsto x(n-x)$ is strictly increasing on integers $0 \leq x \leq n/2$. Because

$$(s(n-s)) \;>\; (d-1)(n-d+1),$$

we must have $s \geq d$. Hence Corollary 2.3 gives

$$\mathsf{s}(G) = \frac{1}{2}(H_s - 1) \geq \frac{1}{2}(H_d - 1) = B_d,$$

with equality exactly when $s = d$, i.e., $G \cong K_{d,n-d}$.

Suppose now that $r \geq 3$. Every edge has at least one endpoint outside the largest class, so $e(G) \leq sn$. We claim that $2s \geq d$. If not, then $s \leq (d-1)/2$, yielding

$$e(G) \leq sn \leq \frac{(d-1)n}{2} < (d-1)(n-d+1),$$

because $n \geq 2d$ implies $n - d + 1 > n/2$. This contradicts (2). Therefore $2s \geq d$. Set $L = \min(n, 2s)$. Since $n \geq 2d$, it follows that $L \geq d$. Moreover $s \geq 2$, hence $L \geq 4$ and $H_L \geq H_4 = 25/12 > 2$. Using Corollary 2.3,

$$\mathsf{s}(G) = H_L - \frac{3}{2} > \frac{1}{2}(H_L - 1) \geq \frac{1}{2}(H_d - 1) = B_d.$$

Thus equality cannot occur when $r \geq 3$.

Finally, the threshold is sharp: $K_{d-1,n-d+1}$ has $(d-1)(n-d+1)$ edges and

$$\mathsf{s}(K_{d-1,n-d+1}) = B_{d-1} < B_d,$$

where $B_1 = 0$ for $d = 2$. This completes the proof. □

Corollary 3.1 (The conjectured edge threshold inside the class)**.** *If G is complete multipartite, $n \geq 2d$, and $e(G) \geq d(n-d)$, then $\mathsf{s}(G) \geq B_d$, with equality only for $K_{d,n-d}$.*

Proof. For $n \geq 2d$,

$$d(n-d) - (d-1)(n-d+1) = n - 2d + 1 > 0.$$

Apply Theorem 1.1. □

4 A bounded vertex-cover extension

Proof of Theorem 1.2. Let S be a vertex cover of minimum size $q \leq d$ and put $T = V(G) \setminus S$. The set T is independent, whence

$$e(G) \leq q(n-q) + \binom{q}{2} = qn - \frac{q(q+1)}{2}. \tag{5}$$

The right side is increasing for integral $0 \leq q \leq d$ under $n \geq 2d$. If $q \leq d-1$, then

$$e(G) \leq (d-1)n - \frac{d(d-1)}{2} < d(n-d)$$

where the strict inequality follows from $n \geq 2d + \binom{d}{2} > \frac{d(d+1)}{2}$. This contradicts our edge hypothesis, so $q = d$.

Write $Y = V(G) \setminus S$, and let $r = e(G[S])$. Let t count the edges missing between S and Y. The two independent- and cross-edge contributions give

$$e(G) = d(n-d) - t + r \geq d(n-d), \quad \text{so} \quad 0 \leq t \leq r \leq \binom{d}{2}.$$

At most t vertices of Y are incident with a missing cross-edge. Consequently at least

$$(n-d) - t \geq n - d - \binom{d}{2} \geq d$$

vertices of Y are adjacent to every vertex of S. These vertices and S contain $K_{d,d}$, which contributes every even cycle length $4, 6, \ldots, 2d$. Hence $\mathsf{s}(G) \geq B_d$.

If $r > 0$, an edge inside S, together with any vertex of Y adjacent to all of S, forms a triangle. This gives the additional length three and strictly increases $\mathsf{s}(G)$. If $r = 0$, then $t = 0$; as Y is independent, the graph is exactly $K_{d,n-d}$. The converse equality is immediate. □

Remark 4.1. *The sufficient size condition in Theorem 1.2 is not asserted optimal. Its conclusion applies to graphs with arbitrary edges inside S and nonuniform neighborhoods in Y, so it does not follow merely from Theorem 1.1.*

5 A near-balanced theorem for arbitrary graphs

We give the external results used in this section in the precise forms needed below. Brandt's theorem [4], also recorded in [1, Theorem 2.5], states that a nonbipartite n-vertex graph with more than $(n-1)^2/4+1$ edges has cycles of every length from 3 to its circumference. Erdős–Gallai [5] shows that a graph with m edges and n vertices has a cycle of length at least $2m/(n-1)$ whenever $m \geq n$. The Moon–Moser bipartite Ore condition [6, Corollary 3] states that a balanced bipartite graph with q vertices in each class is Hamiltonian if every nonadjacent pair x, y in opposite classes satisfies $\deg(x)+\deg(y) > q$.

Lemma 5.1 (Robust balanced bipartite Hamiltonicity)**.** *Let $q \geq 3$. If at most $q-2$ edges are removed from $K_{q,q}$, then the remaining graph has a Hamiltonian cycle.*

Proof. Write $t \leq q-2$ for the number of missing edges. For nonadjacent opposite-class vertices x, y, let m_x, m_y denote the numbers of their missing cross-edges. The absent edge xy is counted twice in m_x+m_y, whereas every other missing edge is counted at most once. Thus $m_x+m_y \leq t+1$, whence

$$\deg(x)+\deg(y) = 2q-(m_x+m_y) \geq 2q-t-1 \geq q+1 > q.$$

The cited Moon–Moser criterion gives a Hamiltonian cycle. □

Lemma 5.2 (Covering four missing cross-edges)**.** *The edge set of any bipartite graph having at most four edges can be covered by at most two vertices from each bipartition class.*

Proof. If some vertex has degree at least three, select that vertex; at most one edge remains uncovered, and one endpoint of that edge completes the cover. If the maximum degree is two, select a vertex of degree two. Of the at most two remaining edges, choose an endpoint on the side opposite the selected vertex for each, using one selected vertex on the first side and at most two on the other. Finally, if every vertex has degree at most one, the edges form a matching of size at most four. Choose an endpoint on the first side of at most two matching edges and an endpoint on the second side of the remaining edges. □

Proof of Theorem 1.3. Write $n = 2d+r$ with $0 \leq r \leq 4$.

Case 1: G is nonbipartite. A direct calculation yields

$$d(n-d) - \left(\frac{(n-1)^2}{4}+1\right) = \frac{4d-4-(r-1)^2}{4} > 0, \tag{6}$$

since $d \geq 4$ and $(r-1)^2 \leq 9$. Brandt's criterion therefore supplies cycles of all lengths from 3 to the circumference $c(G)$. By the Erdős–Gallai estimate,

$$c(G) \geq \frac{2e(G)}{n-1} \geq \frac{2d(n-d)}{n} \geq d.$$

(The last inequality uses $n \geq 2d$.) Hence

$$\mathsf{s}(G) \geq H_d - \frac{3}{2} > \frac{1}{2}(H_d-1) = B_d,$$

because $H_d \geq H_4 = 25/12 > 2$.

Case 2: G is bipartite. Let its bipartition class sizes satisfy $a \le b$, $a+b=n$. Since $e(G) \le ab$ and $e(G) \ge d(n-d)$, the monotonicity of $a(n-a)$ for $a \le n/2$ implies $a \ge d$. Thus

$$a = d+j, \quad b = d+r-j, \quad 0 \le j \le \lfloor r/2 \rfloor.$$

The number t of missing cross-edges satisfies

$$t = ab - e(G) \le j(r-j). \tag{7}$$

If $j=0$, then $t=0$ and $G = K_{d,n-d}$. Otherwise (r,j) belongs to

$$\{(2,1),(3,1),(4,1),(4,2)\},$$

so $t \le 1,2,3,4$, respectively.

We first exhibit a $K_{d,d}$ subgraph. If $j=1$, delete from the larger class an endpoint of each missing edge; this uses at most $t \le r-1$ vertices, leaving at least $(d+r-1)-(r-1) = d$ there and all $d+1$ vertices on the other side. Choose d in each class. If $(r,j)=(4,2)$, both original bipartition classes have size $d+2$. Apply Lemma 5.2 to the graph of missing cross-edges, delete its covering vertices (at most two from each class), and choose d surviving vertices from each class. Thus $K_{d,d} \subseteq G$ in every case with $j>0$, and consequently

$$\{4,6,\dots,2d\} \subseteq \mathcal{C}(G).$$

It remains to find an additional cycle length. For $j=1$, choose $d+1$ vertices from each class. The induced balanced bipartite graph is obtained from $K_{d+1,d+1}$ by removing at most $t \le r-1 \le 3$ edges. Set $q = d+1 \ge 5$; then $t \le 3 \le q-2$, so Lemma 5.1 produces a cycle of length $2d+2$. For $(r,j)=(4,2)$, the entire graph has classes of size $q = d+2 \ge 6$ and at most four missing edges. As $4 \le q-2$, Lemma 5.1 supplies a cycle of length $2d+4$. Both lengths exceed $2d$; the harmonic sum is therefore strictly larger than B_d. The sole equality case is $j=0$, namely $K_{d,n-d}$. □

6 A reduction and the remaining regime

The elementary observation below is useful when searching for a minimal counterexample to (1) at fixed d.

Proposition 6.1 (Deletion of a low-degree vertex)**.** *Fix $d \ge 2$. Suppose $n > 2d$, G has n vertices, $e(G) \ge d(n-d)$, and G has a vertex v with $\deg_G(v) \le d$. Then $G-v$ has $n-1 \ge 2d$ vertices, $e(G-v) \ge d((n-1)-d)$, and $\mathsf{s}(G-v) \le \mathsf{s}(G)$.*

Proof. Removing v deletes at most d edges and cannot introduce new cycle lengths. □

Thus a minimum-order counterexample to the inequality (at a fixed d, if one exists) must either have order $2d$ or have minimum degree at least $d+1$.

The results here do not cover arbitrary graphs of unbounded order for each fixed small d. In particular, neither their correctness nor the exact finite checks below constitutes a solution of the unrestricted Erdős problem. The complete multipartite cycle spectrum is elementary; whether the weighted consequence already appears implicitly in published literature requires additional specialist review. We therefore make no priority claim.

Computational verification (limited scope)

The supplemental program `verify_graph_atlas.py` enumerates all isomorphism types of simple graphs on at most seven vertices using NetworkX's Graph Atlas. It computes the cycle-length spectrum and reciprocal sum with exact rational arithmetic, then compares against B_d for all admissible $2 \le d \le n/2$ and edge thresholds. A local execution found 1,194 qualifying graph–parameter tests and no violations; the only equality isomorphism type in each parameter cell is $K_{d,n-d}$. These finite tests do not directly cover Theorem 1.3, which starts at $n = 8$, and do not replace its proof.

Research status and attribution

This manuscript establishes precisely the restricted theorems stated above, conditional only on the cited standard theorems of graph theory. It does *not* settle Erdős Problem #65. The literature search for this draft was targeted rather than exhaustive; the claimed special-case statements have not been independently certified as new or peer-reviewed. The author placeholder must be replaced by accurate authorship and affiliation information before public release.

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